ROHIT GUPTA

Data Analyst | AI/ML Engineer

+91-9973556808 | Bengaluru, India | rohit.gupta1604004@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Science graduate with published deep learning research (IEEE) and industry experience spanning data analytics and machine learning at Viasat India and IIT Gandhinagar. Skilled in building production-style RAG and multi-agent systems with LangChain, LangGraph, and FAISS, and in translating large-scale data into measurable outcomes across analytics, data science, and applied ML roles.

TECHNICAL SKILLS

Languages: Python, SQL, C++

Machine Learning & Deep Learning: TensorFlow, Keras, Scikit-learn, XGBoost, CNN, LSTM, Transfer Learning, Attention Mechanisms, Hyperparameter Tuning, Model Evaluation

Generative AI & NLP: LangChain, LangGraph, RAG Pipelines, FAISS, Pinecone, LLMs, Embeddings, Prompt Engineering, Knowledge Retrieval

Data & Analytics: Pandas, NumPy, Feature Engineering, EDA, Statistical Analysis, Predictive Modeling

Visualization & Tooling: Tableau, Matplotlib, Seaborn, Git, FastAPI

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Data & Analytics / ***Viasat India*** Feb 2026 – Apr 2026

- Architected an end-to-end analytics pipeline integrating 5+ disparate data sources into a unified, analysis-ready customer dataset, engineered for extensibility so new use cases could be onboarded by referencing existing data without modifying the core pipeline.
- Conducted a **SWOT analysis** on customer segments to inform a segmentation, scoring, and ranking strategy, then formulated the underlying models through feature engineering and normalized metrics to prioritize high-value accounts.
- Delivered interactive **Tableau dashboards** adopted by business stakeholders to support targeted, self-serve outreach decisions.

Summer Research Intern — Two Higgs Doublet Model Exploration / ***IIT Gandhinagar*** May 2024 – Jul 2024

- Engineered and processed 25 high-dimensional datasets (500K rows each) for particle physics parameter-constraint simulations, applying data balancing and feature engineering on imbalanced data.
- Constructed a regularized neural network with dropout, achieving a 15% accuracy improvement over baseline models in parameter constraint prediction.
- Authored research reports with statistical visualizations, contributing to peer-reviewed domain research in data-driven particle physics.

PUBLICATION

COVID Image Classification using Xception with Attention Mechanism

First Author | 2nd International Conference on Circuits, Power, and Intelligent Systems (CCPIS), 2025 | IEEE

- Developed deep learning pipeline for automated COVID-19 detection using the **SARS-CoV-2 CT Scan dataset**.
- Implemented an **Xception-based transfer learning** architecture enhanced with **Multi-Head Channel Attention** (MHCA) for improved feature representation.
- Applied preprocessing, normalization, augmentation, and fine-tuning strategies to improve robustness and reduce overfitting.
- Achieved 97.86% accuracy, 98.91% recall, and 96.82% precision on unseen CT scans.
- Published as **First Author** in an IEEE conference.

PROJECTS

InsightFlow AI — Multi-Agent Analytics Platform | *Python, LangGraph, LangChain, Pandas, Streamlit, LLMs [Live]*

- Built a multi-agent Generative AI platform using LangGraph and LangChain that automates data analysis, visualization, report generation, and natural language querying through specialized AI workflows.
- Implemented **9 specialized AI agents** for data cleaning, statistical analysis, chart explanation, recommendation generation, conversational analytics, and report writing.
- Enabled natural language querying and custom visualization generation, allowing non-technical users to explore data without writing code.

Personalized AI Assistant — RAG-Based Intelligent Agent | *Python, LangChain, FAISS, LLMs [Live]*

- Developed a production-style RAG agent integrating multi-source personal data using embeddings, a FAISS vector store, and an LLM backend to deliver accurate, context-aware responses.
- Structured a modular ingestion pipeline supporting PDF, text, and structured data sources; implemented semantic chunking and retrieval re-ranking to improve answer relevance.

EDUCATION

B.Tech in Computer Science and Engineering 2022 – 2026

Kalinga Institute of Industrial Technology, Bhubaneswar, Odisha, India | CGPA: 8.96

ACHIEVEMENTS

- First-author IEEE publication at the undergraduate level.
- Completed the Agile Virtual Experience Program (JPMorgan Chase & Co.), gaining exposure to agile workflows and cross-functional collaboration.